

STUDENT GOAL TRACKER

Python Project Report

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Academic Project Report

1. Introduction

Student Goal Tracker is a Python-based program designed to help students organize and monitor their academic goals. The program allows a user to enter a goal, associate it with a subject, set a deadline, choose a priority, and track progress. Goal information is stored in a JSON file so that it can be loaded again when the program is started.

2. Problem Statement

Students often have several assignments, examinations, projects, and personal targets to manage. Remembering the deadlines and current progress of all these tasks can be difficult. The project aims to provide a simple command-line system that helps students record goals and monitor their progress.

3. Objectives

- Create a simple goal-management system for students.
- Allow users to add and store academic goals.
- Assign a subject, deadline, and priority to each goal.
- Validate deadlines and prevent past dates from being accepted.
- Track progress for individual goals.
- Generate an overall progress report.
- Use JSON file storage to preserve goal information.

4. Main Features

Feature	Description
Add Goal	Adds a goal with subject, deadline, priority, and initial progress.
View Goals	Displays stored goals and their progress/status.
Update Progress	Allows the user to update the progress of a selected goal.
Progress Report	Shows total goals, completed goals, and overall progress.
Date Validation	Checks for empty, incorrectly formatted, and past deadlines.
JSON Storage	Stores goal information in DATA.json.

5. Technologies and Python Concepts Used

- Python programming language
- JSON for data storage
- datetime module for date handling and validation

- Lists and dictionaries for storing goal information
- Functions for modular program design
- Loops and conditional statements
- Exception handling for invalid input
- File handling for reading and writing DATA.json

6. Working of the Program

Start: The program attempts to load previously stored goals from DATA.json.

Menu: The user chooses Add Goal, View Goals, Update Progress, Progress Report, or Exit.

Add Goal: The user enters goal details. The deadline is checked for format and validity, and a priority is selected.

Store Data: The goal is represented as a dictionary and added to the goals list, then saved to the JSON file.

View: Stored goals are displayed with their subject, deadline, priority, progress, and status.

Progress: The user can select a goal and enter its progress percentage.

Report: The program calculates and displays the number of goals, completed goals, and overall progress.

7. Data Representation

Each goal is stored as a dictionary containing the following fields:

Field	Purpose
goal	Stores the goal description.
subject	Stores the subject associated with the goal.
deadline	Stores the selected deadline.
priority	Stores High, Medium, or Low priority.
progress	Stores progress as a percentage from 0 to 100.

8. Sample Program Interaction

===== STUDENT GOAL TRACKER =====

1. Add Goal
2. View Goals
3. Update Progress
4. Progress Report
5. Exit

Enter your choice: 1

Enter your goal: Complete Calculus Module 3

Enter subject name: Calculus

Enter a deadline like (dd-mm-yyyy): 30-09-2026

Choose the priority of the task:

1. High
2. Medium
3. Low

9. Learning Outcomes

- Understanding how functions can divide a program into manageable tasks.
- Working with lists and dictionaries to organize structured information.
- Reading and writing JSON data using Python.
- Handling invalid user input with exception handling.
- Using datetime for practical date validation.
- Applying Python concepts to a real-world student problem.

10. Future Scope

- Add options to edit and delete existing goals.
- Add automatic deadline reminders.
- Sort goals by deadline, subject, or priority.
- Add a graphical user interface.
- Add charts and more detailed progress statistics.
- Use a database for larger-scale data storage.

11. Conclusion

The Student Goal Tracker demonstrates how Python can be used to solve a practical problem faced by students. It combines functions, lists, dictionaries, loops, conditional statements, exception handling, date validation, and JSON file handling in a single application. The project provides a simple way to record goals, organize them by subject and priority, and monitor progress. It also provides a foundation for future improvements such as reminders, graphical interfaces, and detailed progress analysis.